SONAKSHI CHOWGULE- CNL ASSIGNMENT 02

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter a binary number: ");
        String input = scanner.nextLine();
        boolean isValid = !input.isEmpty();
        int bitCount = 0;

        for (int i=0; i<input.length(); i++) {
            char ch=input.charAt(i);
            if (ch=='0' || ch=='1') {
                bitCount++;
            } else{
                isValid = false;
                break;
            }
        }

        if (isValid) {
            System.out.println("Binary Input: "+input);
            System.out.println("Count(m): "+bitCount);
            int r = 0;
            while(Math.pow(2,r) < bitCount+r+1){
                r++;
            }
            System.out.println("Parity Bits(r): " +r);
            int totalLength = bitCount + r;
            System.out.println("Total length: "+totalLength);
            int[] arr = new int[totalLength+1];

            int dataIndex = 0;
            for(int i=1; i<=totalLength; i++){
                if((i&(i-1)) !=0){
                    arr[i] = Character.getNumericValue(input.charAt(dataIndex));
                    dataIndex++;
                }
            }
            for (int i=0; i<r; i++) {
                int parityPos = (int) Math.pow(2, i);
```

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                int parityValue = 0;

                for (int j = 1; j <= totalLength; j++) {
                    if ((j & parityPos) != 0) {
                        parityValue ^= arr[j];
                    }
            }
            arr[parityPos] = parityValue;
            }

            System.out.print("Generated Hamming Code: ");
            for (int i = 1; i <= totalLength; i++) {
                System.out.print(arr[i]);
            }
            System.out.println();

        }
        else {
            System.out.println("Error: Contains non-binary characters.");
        }

        scanner.close();
    }
}
```

OUTPUT:
Enter a binary number: 1011
Binary Input: 1011
Count(m): 4
Parity Bits(r): 3
Total length: 7
Generated Hamming Code: 0110011

GITHUB LINK:
https://github.com/chowgulesonakshi-lab/Computer-Networks/edit/main/Hamming_Code